

Prismatic Battery Cooling System Individual Report

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Abstract

This study examines the thermal behaviour of a prismatic battery module and evaluates several strategies to enhance its cooling effectiveness. A discretisation comparison showed that the High Resolution scheme more accurately captured temperature gradients and hotspot locations than first-order upwind, and was therefore adopted for all subsequent simulations. A parametric spacing study was then performed using 25 mm, 50 mm and 65 mm cell gaps. The 65 mm spacing produced the lowest average surface temperature and the smoothest mid-plane distribution, improving the average temperature by approximately 4.1K relative to the 50 mm baseline, and was thus selected for optimisation.

Three geometric modifications to the 65mm spaced cell were investigated to address the persistent bypass flow and uneven cooling observed in the baseline: a 10 mm coolant channel surrounding each cell, inter-cell baffles to redirect flow, and a combined design integrating both features. Mid-plane contours, surface temperatures and streamline patterns showed that the combined configuration delivered the most substantial improvement. Compared with the original 50 mm baseline, the combined design further reduced the average surface temperature from 328.2K to 301.8 K, a dramatic 26.4 K reduction, and achieved far more uniform flow distribution. These results highlight the strong impact of targeted flow control on battery thermal management and identify the combined modification as the most effective cooling strategy.

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1 Discretisation Schemes

ANSYS CFX provides two main advection discretisation schemes: the first-order upwind scheme and the High Resolution scheme. The group report adopted High Resolution, and both approaches are briefly evaluated here to understand their influence on the predicted thermal field.

The first-order upwind scheme is unconditionally bounded and highly stable, but introduces substantial numerical diffusion due to its first-order accuracy [1, 2]. This smoothing effect suppresses local gradients and generally underpredicts peak temperatures. In contrast, the High Resolution scheme used by CFX is a bounded second-order method that locally reduces to upwind only where required to maintain stability [3]. This allows it to capture steeper gradients and inter-cell hotspots with reduced artificial diffusion.

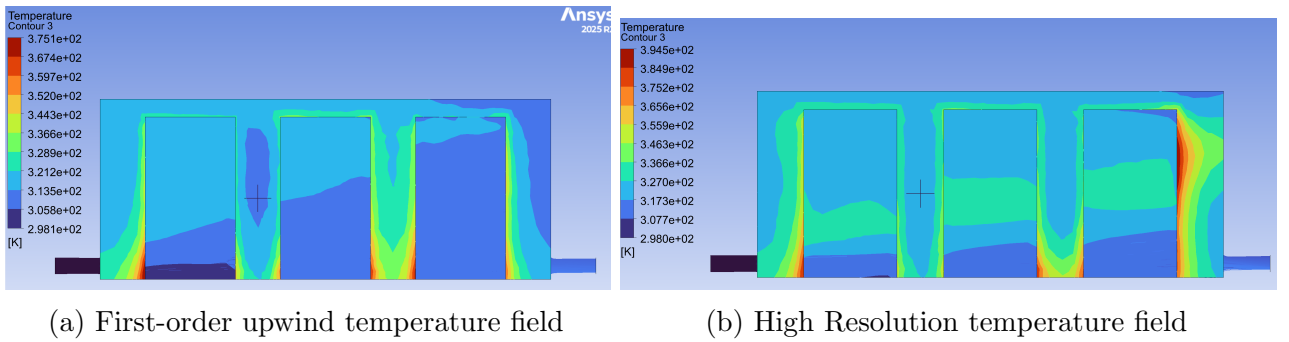


Figure 1: Mid-plane temperature distributions for the two schemes.

To quantify these differences, temperatures at four monitoring points (Z1–Z4) along the mid-plane were extracted for both schemes. Table 1 shows that High Resolution consistently predicts higher local temperatures at Z1, Z2, and Z4, reflecting sharper gradients and reduced diffusion. The lower numerical diffusion also reveals a more distinct hotspot near Z4, where coolant velocity is lowest.

Table 1: Mid-plane temperature comparison at monitoring points Z1–Z4.

Point	First-order	High Res.	ΔT [K]
Z1	319.2	326.8	+7.6
Z2	313.4	322.9	+9.5
Z3	322.2	320.1	-2.1
Z4	314.2	336.0	+21.8

Behaviour of the Two Schemes

The contrasting thermal fields in Fig. 1 and the pointwise values in Table 1 reflect the underlying behaviour of the two advection schemes.

First-order upwind: This scheme computes the transported quantity using only upstream values, e.g.

$$\phi_f = \phi_{\text{upstream}},$$

which introduces first-order truncation error and significant numerical diffusion [1, 2]. This diffusion acts like an added artificial conductivity, smoothing temperature gradients and suppressing peaks. This explains the systematically lower temperatures at Z1, Z2 and Z4, and the visibly flatter mid-plane contour in Fig. 1. While highly stable, this scheme cannot resolve inter-cell hotspots accurately.

High Resolution: The High Resolution scheme used by CFX is a bounded second-order method [3]. It behaves like a second-order scheme in smooth regions,

$$\phi_f = \phi_{\text{HR}}(\nabla\phi),$$

and blends locally to first-order only where required to maintain monotonicity. This reduces artificial diffusion and preserves sharper gradients, leading to the higher peak temperatures seen at Z1, Z2 and especially Z4 (+21.8 K). The temperature field in Fig. 1 therefore captures the stronger hotspot near the outlet-side cell that the upwind scheme smooths out.

Overall, the High Resolution scheme captures steeper and more realistic temperature gradients, whereas first-order upwind produces a smoother but less accurate field. Given its improved predictive capability and consistency with the group report, the High Resolution scheme is adopted for all remaining simulations.

2 Parametric Study

2.1 Case Study Outline

This parametric study investigates the spacing between battery cells and its influence on mid-plane temperature distribution, hotspot development, and coolant flow behaviour. Three inter-cell spacings were examined: 25 mm, 50 mm (baseline), and 65 mm. In all configurations, the surface area of each battery generated a constant heat load of 350 W, ensuring that any differences in cooling performance arise solely from geometric effects rather than thermal input. Larger spacings are expected to perform better by allowing greater flow penetration and more uniform coolant distribution between the cells. The results are grouped by relevant physical quantities to enable direct comparison of thermal and hydraulic performance across all spacing configurations.

2.2 Case Study Results

Mid-Plane Temperature Distribution

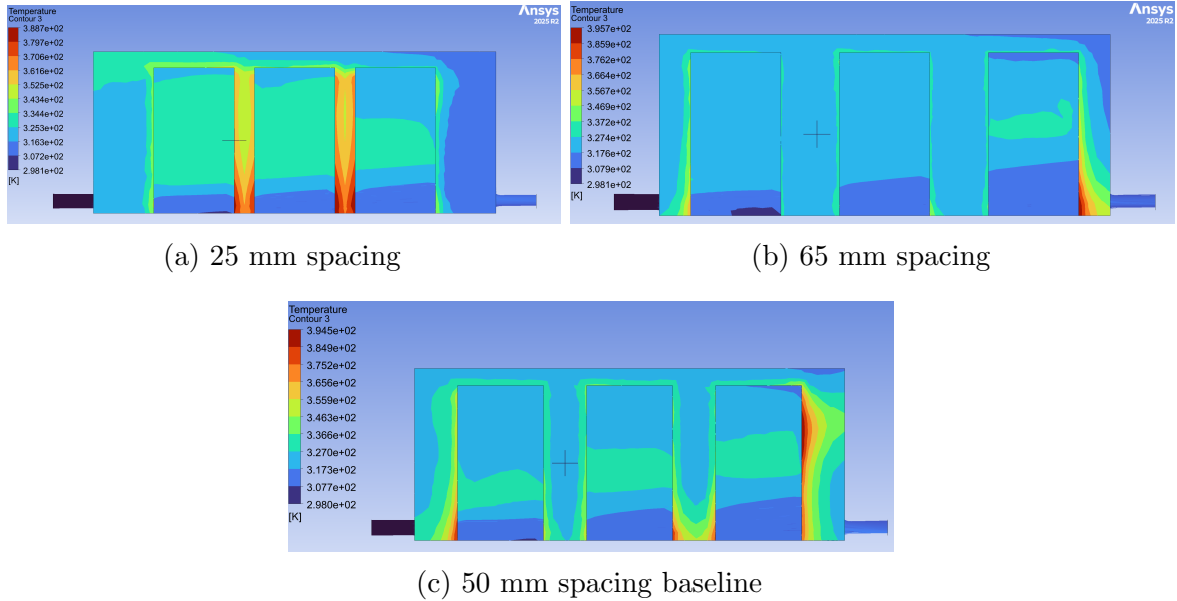


Figure 2: Mid-plane temperature comparison for different cell spacings.

Mid-Plane Temperature Summary: The mid-plane temperatures show that reducing the cell spacing intensifies thermal loading between the batteries. At 25 mm spacing, the coolant is highly restricted, leading to significantly elevated temperatures at Z2 (351.2 K) and Z3 (367.3 K), with steep gradients across the mid-plane. The 50 mm baseline spacing improves coolant penetration, reducing Z2 and Z3 to 322.9 K and 320.1 K respectively, though a clear hotspot still forms at Z4 (336 K). When the spacing is increased to 65 mm, the central zones become more uniform, with Z1–Z3 stabilising around 323–325 K; however, the outlet-side region remains thermally dominant, with Z4 rising to 335.2 K.

Overall Trend As spacing increases, the mid-plane temperature field becomes more uniform and the number of hotspots decreases. Wider channels (65 mm) allow smoother coolant distribution and significantly lower temperatures in the inner regions compared to the 25 mm case. However, larger spacing does not guarantee a lower maximum temperature: the peak shifts to a single outlet-side hotspot (Z4), where recirculation is strongest. Thus, increased spacing improves overall uniformity but can still permit an isolated high-temperature region to develop.

Cell Surface Hotspot Contours

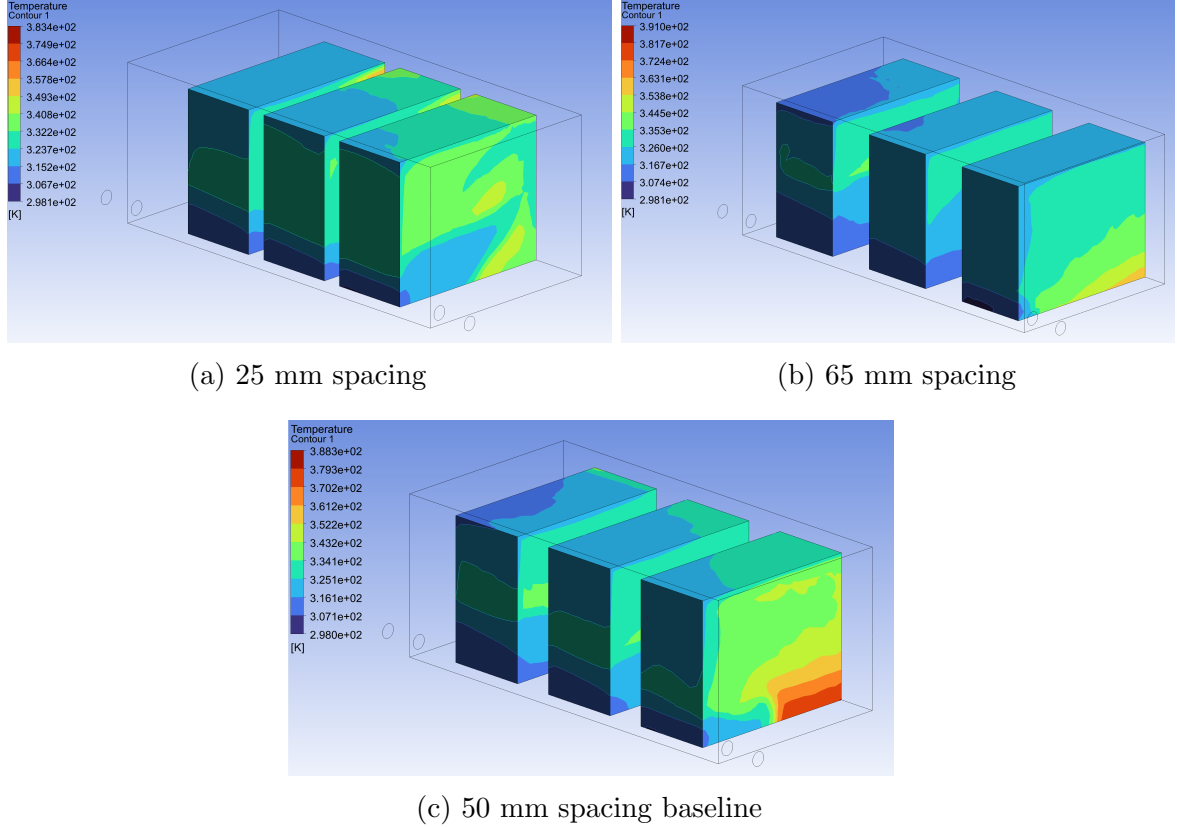


Figure 3: Cell surface temperature contours and hotspot formation across cell spacings.

Cell Contour Behaviour. The surface temperature contours reveal that the number of hotspots decreases as the inter-cell spacing increases, but the *peak* hotspot temperature rises. For the 25 mm spacing, multiple hotspots form due to restricted coolant flow, with a maximum surface temperature of 383.4 K. The 50 mm baseline spacing allows better coolant penetration and reduces the number of hotspots, yet the dominant hotspot increases to 388.3 K, concentrated on the inlet-side cell. At 65 mm spacing, the temperature field becomes noticeably smoother and the inter-cell gradients weaken; however, a single strong recirculation region develops near the outlet, raising the hotspot temperature to 391.0 K. This behaviour suggests that although wider spacing improves flow uniformity between the cells, it can also promote larger low-velocity regions where heat accumulates. Consequently, increasing spacing reduces hotspot count but does not necessarily reduce the maximum surface temperature across the battery pack.

Coolant Flow Streamlines

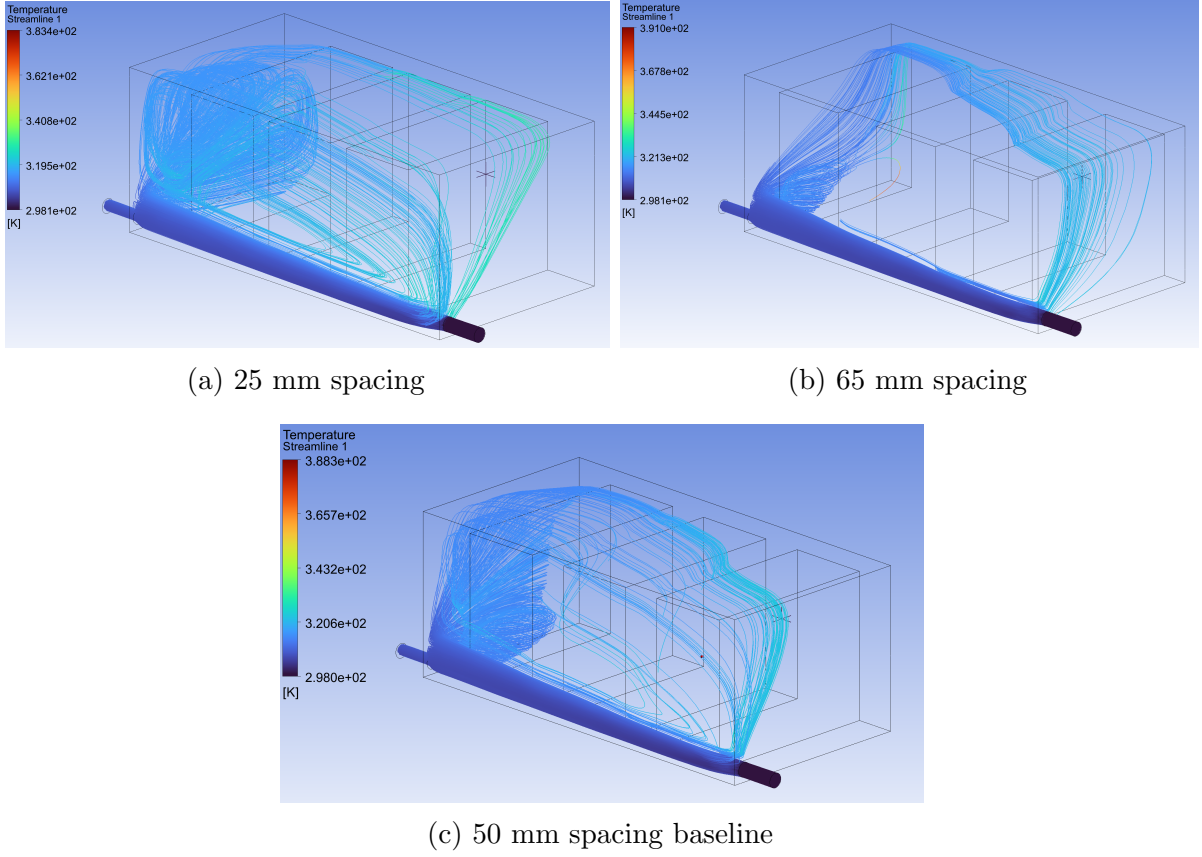


Figure 4: Coolant streamline behaviour for each cell spacing configuration.

Streamline Behaviour. In all spacing configurations, the flow forms a dominant bypass jet that travels directly from inlet to outlet, with only limited penetration into the inter-cell gaps. At 25 mm spacing, the restricted geometry produces a large recirculation region behind the final cell, which traps warm fluid and contributes to the higher temperatures observed there. As spacing increases, this recirculation zone reduces in size; however, the stronger bypass jet at 65 mm spacing leaves a low-velocity pocket near the outlet where heat accumulates. This explains why wider spacing smooths the temperature distribution but still results in a higher outlet-side hotspot.

Summary and Discussion

Table 2: Mid-plane temperatures at monitoring zones Z1–Z4 for each cell spacing configuration. All values in Kelvin (K).

Mid-Plane Zone [K]	25 mm	50 mm (Baseline)	65 mm
Z1	325.1	326.8	323.8
Z2	351.2	322.9	325.4
Z3	367.3	320.1	324.6
Z4	313.6	336.0	335.2
Average Cell Temperature [K]	326.2	328.2	324.1
Maximum Surface Temperature [K]	383.4	388.3	391.0

The average cell temperatures in Table 2 were obtained using the `areaAve(Temperature)` function in CFX-Post, applied to each battery’s external surface. The results demonstrate a clear influence of spacing on both the uniformity and the magnitude of temperatures inside the pack.

At 25 mm spacing, the mid-plane temperatures at Z2 (351.2 K) and Z3 (367.3 K) are the highest, reflecting the restricted coolant penetration and the large recirculation region observed behind the final cell. Increasing the spacing to the 50 mm baseline substantially reduces the mid-plane temperatures in these central regions; however, it unexpectedly produces the highest overall average cell temperature (328.2 K). This is due to a large hotspot forming where the bypass jet impacts the inlet-side cell, limiting local cooling.

The 65 mm spacing yields the most uniform mid-plane distribution, with Z1–Z3 tightly clustered between 323–325 K. It also delivers the lowest average cell temperature (324.1 K), indicating improved coolant coverage and reduced internal thermal gradients. The maximum surface temperature does increase to 391.0 K, caused by a single dominant hotspot near the outlet where a low-velocity pocket forms due to the strengthened bypass jet.

Overall, despite the higher peak temperature, seen near the outlet, the 65 mm spacing provides the most favourable thermal performance, achieving the lowest average cell temperature and the smoothest mid-plane distribution. For these reasons, the 65 mm spacing is selected for all subsequent design and optimisation work.

3 Modified/Optimised Cases Results

Based on the preliminary spacing study, the 65 mm configuration demonstrated the most favourable cooling performance and is therefore adopted as the baseline geometry for further optimisation. Three modifications were examined to improve coolant distribution and heat removal:

- **Cell-channel design** — a 10 mm uniform coolant sheath around each cell to maintain close coolant contact along all surfaces.
- **Baffle design** — slotted baffles placed within the module to prevent coolant bypass and force flow through the inter-cell gaps.
- **Combined design** — integration of both channels and baffles to maximise flow guidance and surface coverage.

These concepts aim to enhance local convective heat transfer, suppress hotspot formation, and reduce temperature non-uniformity across the battery module. Performance will be determined by max and average temperature on the battery cell surface.

3.1 Design Description

Cell-channel design

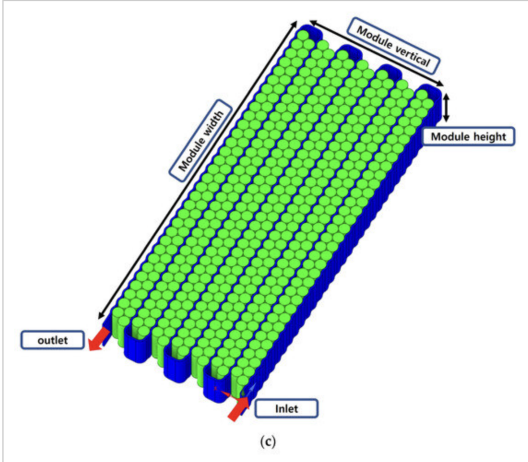
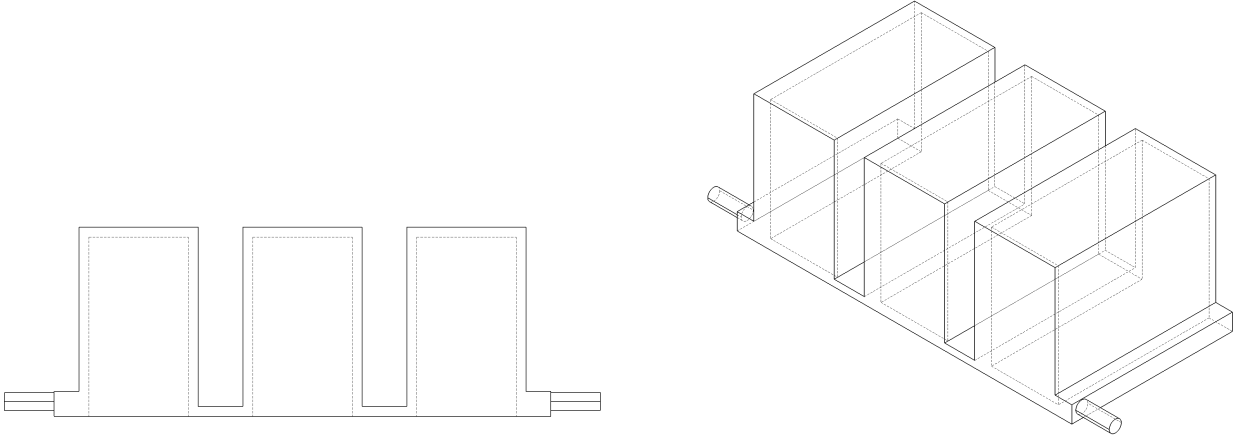


Figure 5: Cooling channel configuration proposed by Jeon et al. (2023).

The cell-channel design in this work is directly inspired by the cooling strategy presented by Jeon et al. [4], who showed that maintaining a directed, continuous coolant path close to the battery surfaces improves heat removal and prevents stagnant regions from forming. Their results demonstrated that channel layouts which keep coolant tightly guided around each cell reduce peak temperatures and enhance temperature uniformity across the module.

A key takeaway from their study is that near-wall cooling—ensuring coolant remains tightly guided around each cell—significantly increases convective heat transfer by minimising separation zones and encouraging uniform flow distribution. This motivated the introduction of a uniform 10 mm coolant gap around each prismatic cell in the present work.

Hypothesis. The proposed channel (Fig. 6) is expected to improve overall cooling by increasing surface wetting and reducing large stagnant pockets observed in the baseline spacing cases. However, due to its unobstructed flow path to the outlet, it may not fully eliminate local hotspots if small recirculation zones remain in low-velocity regions.



(a) Front view of the 10 mm coolant channel.

(b) Isometric view of the channel geometry.

Figure 6: Proposed cell-channel cooling design motivated by near-wall flow strategies.

Baffle design

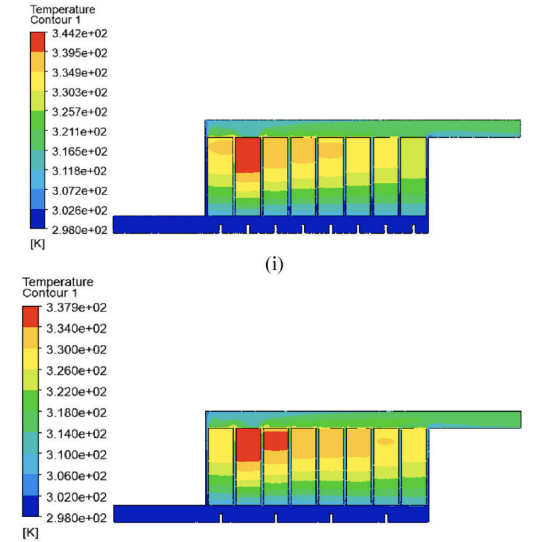
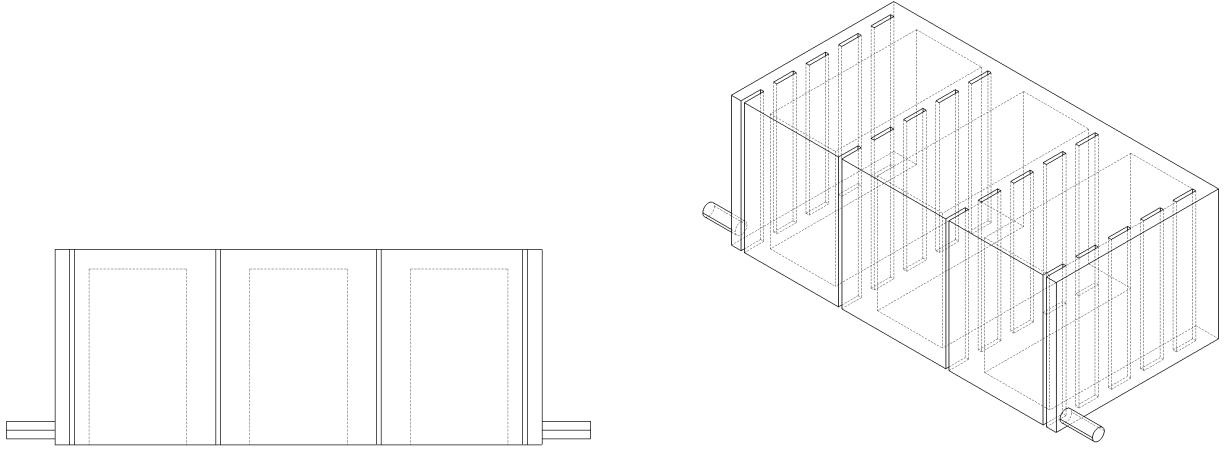


Figure 7: Straight baffle configurations studied by Oyewola et al. (2024).

The baffle design developed in this report adapts the cooling strategy proposed by Oyewola et al. [5], who showed that introducing flow-redirecting baffles into a thermal management system significantly improves cooling uniformity by eliminating low-velocity thermal pockets and forcing the coolant through regions that normally experience poor circulation. In their study, the inclusion of two inclined baffles led to a 1.65 K reduction in maximum temperature and a 1.43 K reduction in temperature non-uniformity ($\Delta T_{\max}$) compared with the no-baffle case. Their optimised geometry—two baffles inclined at 60° —achieved a 2.65 K reduction in peak temperature while increasing pressure drop by only 1.12 Pa, a negligible penalty.

Whereas Oyewola et al. placed baffles upstream in the plenum, this report adapts the concept by positioning perforated baffles directly between the prismatic battery cells. This configuration (Fig. 8) is intended to interfere with the coolant’s natural tendency to bypass the internal region of the pack to the outlet, instead forcing the fluid to weave between cells, increasing turbulence levels, and enhancing inter-cell convective heat transfer.

Hypothesis. Placing baffles between cells should: (i) increase coolant penetration into the inter-cell gaps, (ii) disrupt recirculation pockets that were observed in the baseline 65 mm spacing case, (iii) increase local heat transfer coefficients by accelerating flow through narrow passages, and (iv) reduce both peak temperature and temperature non-uniformity.



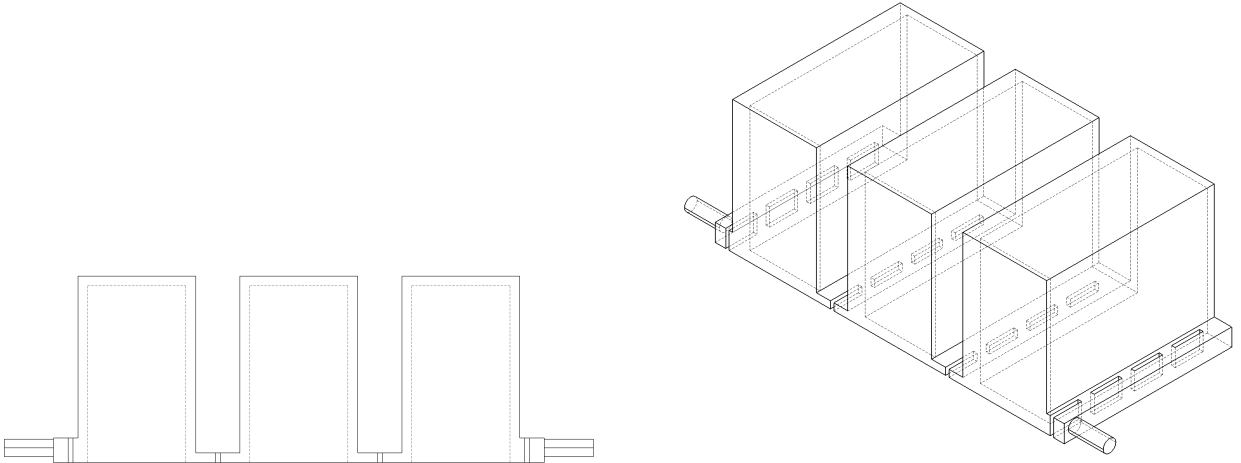
(a) Front view of inter-cell baffle placement.

(b) Isometric view of perforated baffle geometry.

Figure 8: Proposed inter-cell baffle cooling design inspired by Oyewola et al. (2024).

Combined design

The combined design (Fig 9) integrates the strengths of both the cell-channel and baffle concepts. The 10 mm coolant channel keeps flow tightly constrained around the cell surfaces, promoting higher local velocities and reducing the stagnant regions observed in the baseline spacing configuration. The addition of inter-cell baffles prevents coolant from taking a low-resistance bypass path directly toward the outlet, instead redirecting it through the central gaps where cooling demand is highest. In this way, the combined geometry aims to pair the improved surface wetting of the channel design with the enhanced flow guidance and mixing provided by the baffles.



(a) Front view.

(b) Isometric view.

Figure 9: Combined cooling design integrating both constrained flow passages and directional flow control.

3.2 Optimisation Results

Mid-Plane Temperature Distribution

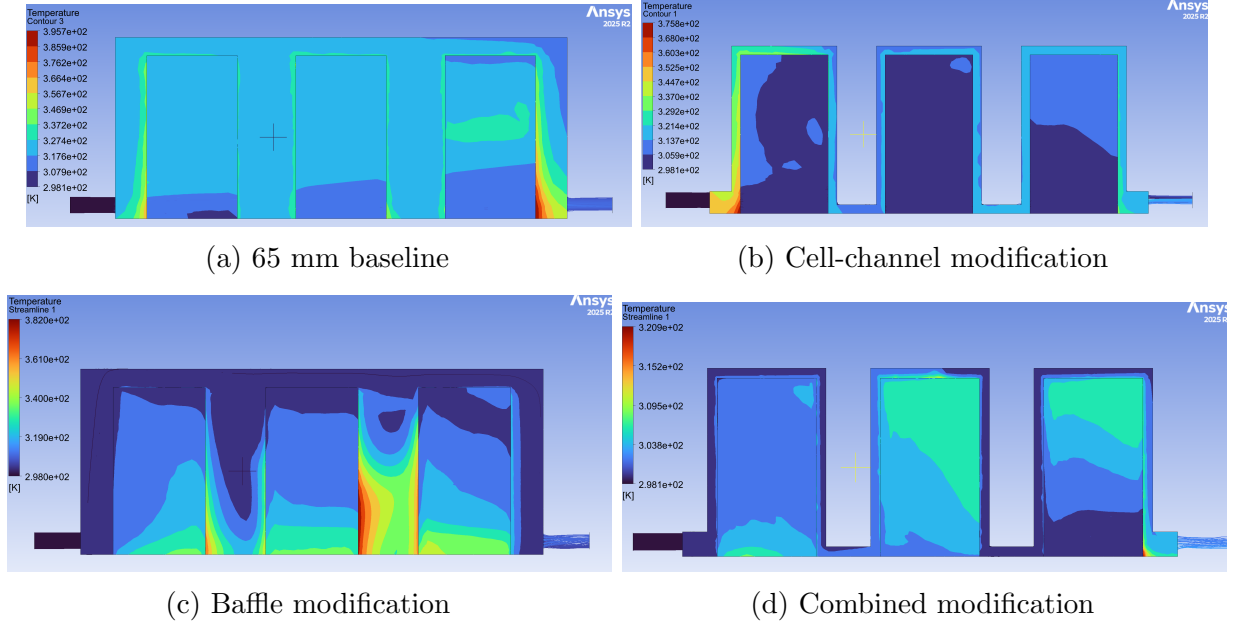


Figure 10: Comparison of mid-plane temperature distribution for baseline and modified designs.

The cell-channel modification reduces the large hotspot near the outlet of the baseline case by keeping coolant closer to the cell surfaces, but the unobstructed longitudinal passage still allows some bypass. This now produces a local hotspot near the inlet-side cells, although the rest of the mid-plane is noticeably smoother than the baseline.

The baffle modification is more ambiguous. Intuitively, forcing coolant through the gaps between cells should enhance cooling, yet the mid-plane field still exhibits relatively high temperatures between the second and third cells. This suggests a more complex interaction between flow redirection and local heating, so the baffle case needs to be assessed together with the cell-surface contours and streamlines before drawing firm conclusions.

The combined modification delivers the most uniform mid-plane distribution. By constraining the flow near the cell surfaces and simultaneously preventing a direct bypass path, it removes both the broad hotspot of the baffle and the local hotspots seen in the other designs. Only a small warm region remains near the outlet, making the combined configuration the most promising from a mid-plane temperature perspective.

Cell Surface Hotspot Contours

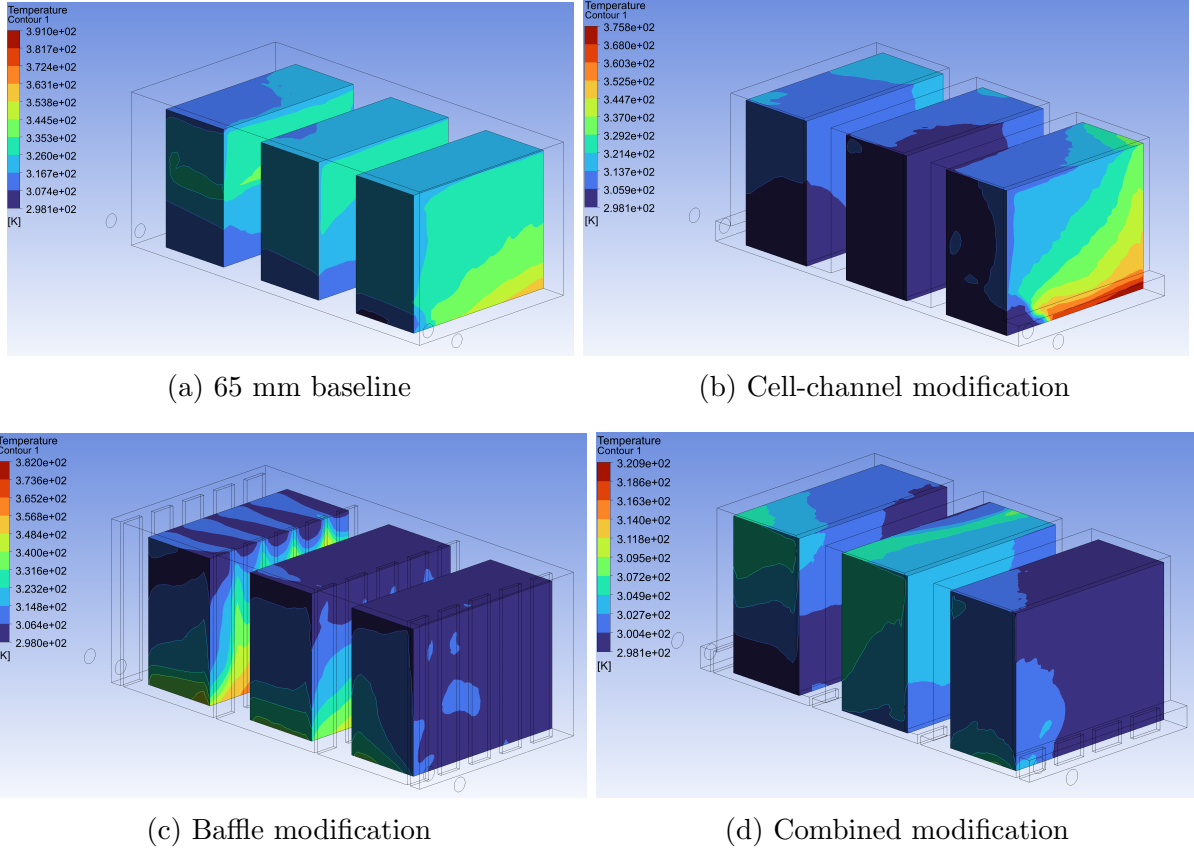


Figure 11: Cell surface temperature contour comparison for baseline and modified designs.

In the cell-channel modification, the hotspot becomes even more pronounced. The first cell face receives very limited coolant contact as the main flow path shifts away and bypasses this region entirely. Compared to the baseline, the channel design smooths temperatures further downstream but clearly sacrifices cooling on the inlet-side cell surface, producing a sharper and more localised hotspot.

This view of the baffle modification helps explain the mid-plane behaviour: although baffles increase mixing, the contours show that coolant is diverted upward, leaving the lower regions of the cells under-cooled. This creates tall vertical temperature bands and confirms that the flow “steps over” parts of the geometry rather than uniformly sweeping along the entire cell height.

The combined modification performs noticeably better than all other cases. The entire cell surface remains within a narrow range of approximately 298–321 K, whereas the baseline, channel, and baffle cases reach temperatures near 380 K. The combined design eliminates the large inlet-side and mid-pack hotspots, indicating that the guided flow path, combined with the baffles for mixing, forces coolant to contact all major heat-generating surfaces more effectively. This strong reduction in both peak temperature and surface temperature variation highlights the combined configuration as the most thermally effective of the tested designs.

Coolant Flow Streamlines

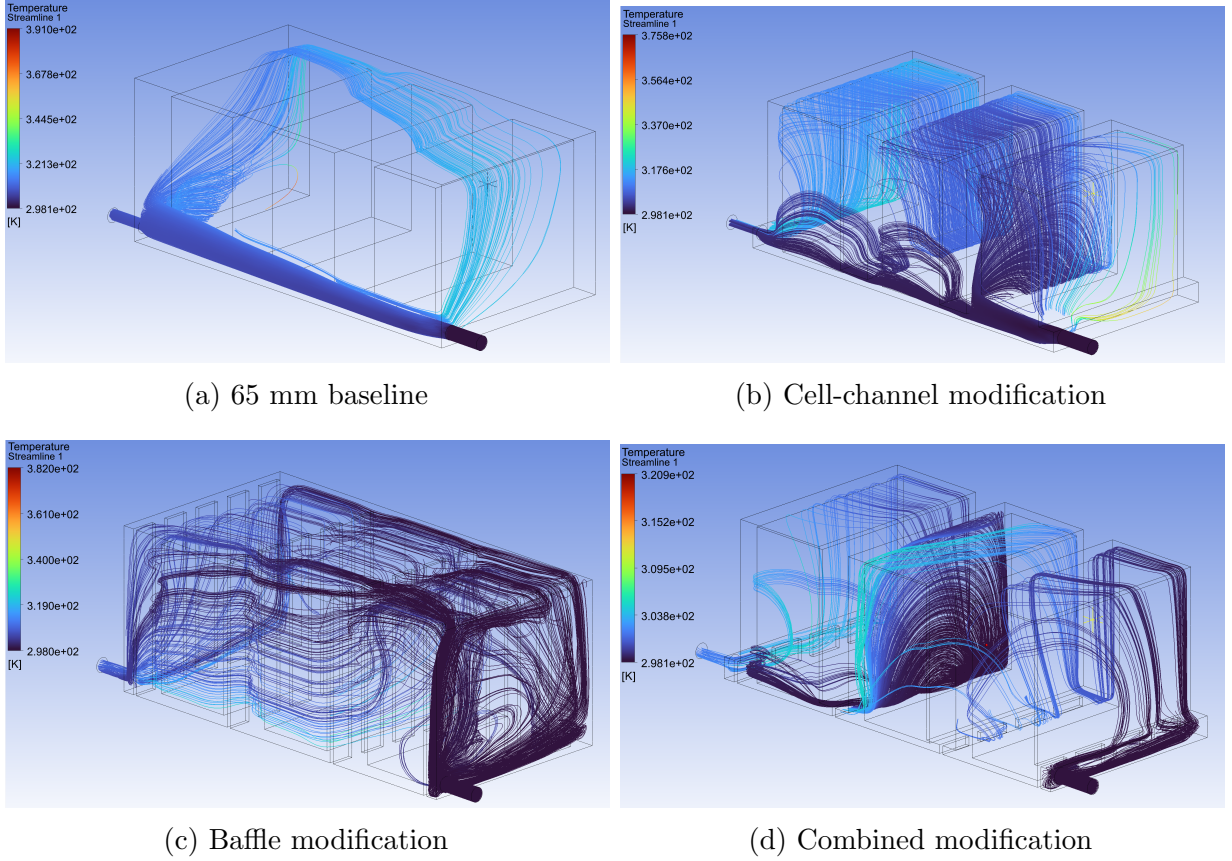


Figure 12: Streamline comparison between baseline and modified designs.

The baseline and cell-channel cases both show a strong jet bypassing the central region of the pack, with only limited coolant entering the inter-cell gaps. The channel modification improves mixing downstream of the first cell, but the initial bypass still leaves the inlet-side surfaces under-cooled.

The baffle modification removes the bypass entirely and drives coolant through the gaps, creating far stronger mixing; however, the increased resistance near the lower regions causes much of the flow to rise and travel over the top surfaces, leaving some lower areas with reduced velocity.

The combined modification provides the most balanced flow behaviour. The channel geometry forces coolant around the cells, while the baffles break up the flow and redistribute it, preventing any direct jetting. This results in broadly uniform coverage of the cell surfaces and no meaningful bypass. Although the streamline rendering contains a minor visual error, the velocity vectors and temperature contours confirm the uniform flow distribution. The overall trend clearly indicates that the combined configuration achieves the most effective flow distribution.

Summary and Discussion

Table 3: Cooling performance comparison across all design configurations.

Metric		Baseline (65 mm)	Cell- channel	Baffle	Combined
Average	Surface	324.1	310.9	318.2	301.8
Temp [K]					
Max	Surface	391	380.0	382.2	320.9
Temp [K]					

As shown in Table 3, all three modifications improve cooling performance relative to the 65 mm baseline, but the combined design delivers the most substantial gains. The cell-channel configuration lowers both the average and peak surface temperatures compared with baseline, although its bypass-induced hotspot limits its effectiveness. The baffle design achieves moderate improvements and removes the main jet bypass, yet still exhibits uneven cooling where flow is redirected over the cell tops. In contrast, the combined modification produces the lowest average temperature (301.8 K) and a dramatically reduced peak temperature of only 320.9 K, far below the 380–390 K range observed in all other cases. Overall, the combined design was selected as the final optimised configuration because it simultaneously prevented bypass, enhanced mixing, and maintained near-wall flow, achieving the lowest temperatures across all metrics.

4 Conclusions

The discretisation study showed that the High Resolution scheme provided sharper gradients and more reliable hotspot prediction than first-order upwind, and was therefore kept for the remainder of the work. The subsequent parametric study demonstrated that cell spacing strongly affects the magnitude and uniformity of cell temperatures. As reported in Table 2, the average cell temperatures were 326.2 K for 25 mm, 328.2 K for the 50 mm baseline, and 324.1 K for 65 mm spacing, meaning the 65 mm configuration reduced the average temperature by 2.1 K relative to 25 mm and 4.1 K relative to 50 mm. Although the 65 mm case exhibited the highest maximum surface temperature (391.0 K) due to a single outlet-side hotspot, it provided the smoothest mid-plane distribution and the lowest overall average cell temperature, and was therefore selected as the baseline for optimisation.

Building on this baseline, three design variations were assessed to address the persistent bypass flow and outlet-side hotspot. The cell-channel design improved downstream cooling but introduced a new inlet-side hotspot due to flow detachment. The baffle design eliminated the main bypass and increased mixing, yet caused uneven cooling where the redirected flow travelled over the cell tops. The combined configuration successfully integrated the strengths of both approaches, producing uniform mid-plane contours, smooth cell-surface temperatures, and balanced streamlines with no meaningful bypass. This design achieved the lowest average cell temperature (301.8 K) and a markedly reduced peak temperature (320.9 K), outperforming all

other configurations.

Overall, the study demonstrates that targeted control of coolant pathways is essential for mitigating thermal gradients within prismatic battery packs. The combined modification offers the most effective and practical improvement, providing a clear path for future refinement of battery thermal management systems.

Acknowledgements

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References

- [1] H. K. Versteeg and W. Malalasekera, *An Introduction to Computational Fluid Dynamics: The Finite Volume Method*, 2nd ed., Pearson, 2007.
- [2] J. H. Ferziger and M. Perić, *Computational Methods for Fluid Dynamics*, 3rd ed., Springer, 2002.
- [3] ANSYS Inc., *ANSYS CFX-Solver Theory Guide*, Release 2024.
- [4] H. Jeon, J. Kim, J. Lee, B. Park, and T. Kim, “Cooling strategy optimization of cylindrical lithium-ion batteries under various charge–discharge conditions,” *Energies*, vol. 16, no. 23, p. 7860, 2023. doi:10.3390/en16237860.
- [5] O. M. Oyewola, M. Sadek, A. A. Abidoye, and F. I. Owuna, “Influence of straight and inclined baffles on enhancement of heat transfer in a lithium–ion battery thermal management system,” *Batteries*, vol. 10, no. 5, p. 125, 2024. doi:10.3390/batteries10050125.